

INÉS PASTOR-MARAZUELA

Postdoctoral Researcher – ASTRON, University of Amsterdam

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EXPERIENCE

Postdoctoral Researcher ASTRON, UvA

Apr 2025 – Present The Netherlands

Transient searches in the image domain on minute timescales with the AstroFlash team. Searching **LOFAR/LoTSS** data, in preparation for the **EuroFlash** project with LOFAR 2.0.

Rubicon Research Fellow JBCA, University of Manchester

Jan 2023 – Mar 2025 Manchester, United Kingdom

Fast Radio Burst studies within the MeerTRAP project using **MeerKAT**. Localisation, high spectro-temporal resolution, polarisation analyses.

Postdoctoral Researcher ASTRON

Sept – Dec 2022 (4 months) Dwingeloo, The Netherlands

Fast Radio Burst studies using data collected during the Apertif survey.

PhD API, University of Amsterdam

2018 – 2022 (48 months) Amsterdam, the Netherlands

Fast Radio Bursts searches and follow-up observations with **Apertif**. FRB triggered searches with **LOFAR**. Observations at the William Herschel Telescope in the optical and infrared.

Master thesis IRAP, Université Paul Sabatier

Jan. – June 2018 (6 months) Toulouse, France

Development of the "EXOD" algorithm to search for transients in **XMM-Newton** data.

EDUCATION

Master in Astrophysics (ASEP) University Paul Sabatier

2016 – 2018 Toulouse, France

Graduated with honors.

Bachelor in Physics Université Paul Sabatier

2013 – 2016 Toulouse, France

Graduated with honors.

TEACHING EXPERIENCE

ASPIRE supervisor

Summer 2025 Anton Pannekoek Observatory

Supervising a student from Ethiopia within ASPIRE on transient searches in the LOFAR/LoTSS survey.

Astronomical Observation Lab

2020 – 2021 Anton Pannekoek Observatory

Teaching assistant to 3rd year bachelor students from September 2020 to January 2021.

Individual supervision

Mar – Sept 2020 IRAP, Toulouse, France

Master thesis co-supervisor of master student Florent Castellani.

Structure and Evolution of Stars

2018 – 2020 University of Amsterdam

Teaching assistant to 1st year master students during two years.

PRIZES AND AWARDS

Rubicon Grant

I was awarded the NWO Rubicon grant during the second round of 2022. This grant allows recently graduated researchers from the Netherlands to carry out independent, self-proposed research at a foreign institution.

IAU PhD Honourable Mention

The International Astronomical Union (IAU) Division D – *High energy astrophysical phenomena* bestowed their 2022 Honourable Mention on my thesis.

OUTREACH

Outreach UvA

Teaching assistant from September 2019 to August 2020 at the Anton Pannekoek Institute (University of Amsterdam). Participation in monthly stargazing nights and events throughout the year.

Jóvenes Científicos Segovianos

Ráfagas Rápidas de Radio: un misterio extragaláctico.

Outreach talk about my research in my hometown, Segovia (Spain), on May 4th 2022.

Bluedot festival

Fast Radio Bursts: an extragalactic mystery. Talk about my research at the Bluedot festival on July 23rd 2023. The festival combines science and music at the Jodrell Bank Observatory.

SKILLS

Languages

Spanish
English
French
Dutch



PUBLICATIONS

First author journal articles

Inés Pastor-Marazuela, Alexa C Gordon, et al. *et al.* "Localization and host galaxy identification of new fast radio bursts with MeerKAT" *Monthly Notices of the Royal Astronomical Society* 545 4 staf2144, 2026

Inés Pastor-Marazuela "Fast radio bursts and the radio perspective on multi-messenger gravitational lensing" *Philosophical Transactions of the Royal Society of London Series A* 383 p. 20240121, 2025

Inés Pastor-Marazuela, Joeri van Leeuwen, Anna Bilous, Liam Connor, Yogesh Maan, Leon Oostrum, Emily Petroff, Dany Vohl, et al. *et al.* "Comprehensive analysis of the Apertif fast radio burst sample - Similarities with young energetic neutron stars" *Astronomy & Astrophysics* 693 A279, 2025

Inés Pastor-Marazuela "Fast Radio Bursts and the radio perspective on multi-messenger gravitational lensing" , 2024

Inés Pastor-Marazuela, Joeri van Leeuwen, Anna Bilous, Liam Connor, Yogesh Maan, Leon Oostrum, Emily Petroff, Samayra Straal, et al. *et al.* "A fast radio burst with submillisecond quasi-periodic structure" *Astronomy & Astrophysics* 678 A149, 2023

I. Pastor-Marazuela, S. M. Straal, et al. *et al.* "New upper limits on low-frequency radio emission from isolated neutron stars with LOFAR" *Astronomy & Astrophysics* 672 A151, 2023

Inés Pastor-Marazuela, Liam Connor, et al. *et al.* "Chromatic periodic activity down to 120 megahertz in a fast radio burst" *Nature* 596 7873 pp. 505–508, 2021

I. Pastor-Marazuela, N. A. Webb, et al. *et al.* "The EXOD search for faint transients in XMM-Newton observations: Method and discovery of four extragalactic Type I X-ray bursters" *Astronomy & Astrophysics* 640 A124, 2020

Highlighted collaboration journal articles

Manisha Caleb et al. *et al.* *A fast radio burst from the first 3 billion years of the Universe* , 2025

J. Tian, I. Pastor-Marazuela, et al. *et al.* "MeerKAT discovery of a hyperactive repeating fast radio burst source" *Monthly Notices of the Royal Astronomical Society* 540 pp. 1685–1700, 2025

K. M. Rajwade, J. Tian, et al. *et al.* "A Coherent Radio Burst from an X-Ray Neutron Star in the Carina Nebula" *The Astrophysical Journal* 985 p. L3, 2025

K. Y. Hanmer et al. *et al.* "Contemporaneous optical-radio observations of a fast radio burst in a close galaxy pair" *Monthly Notices of the Royal Astronomical Society* 538 pp. 1800–1815, 2025

A. V. Bilous et al. *et al.* "An activity transition in FRB 20201124A: Methodological rigor, detection of frequency-dependent cessation, and a geometric magnetar model" *Astronomy and Astrophysics* 696 A194, 2025

J. Tian, K. M. Rajwade, et al. *et al.* "Detection and localization of the highly active FRB 20240114A with MeerKAT" *Monthly Notices of the Royal Astronomical Society* 533 pp. 3174–3193, 2024

K. M. Rajwade, L. N. Driessen, et al. *et al.* "A study of two FRBs with low polarization fractions localized with the MeerTRAP transient buffer system" *Monthly Notices of the Royal Astronomical Society* 532 pp. 3881–3892, 2024

Joeri van Leeuwen et al. *et al.* "The Apertif Radio Transient System (ARTS): Design, commissioning, data release, and detection of the first five fast radio bursts" *Astronomy and Astrophysics* 672 A117, 2023

L. Connor et al. *et al.* "A bright, high rotation-measure FRB that skewers the M33 halo" *Monthly Notices of the Royal Astronomical Society* 499 pp. 4716–4724, 2020

INVITED TALKS AND COLLOQUIA

FRB2026

Invited talk 📅 July 6 2026 📍 Guiyang, China

Dutch SKA days

Invited talk 📅 March 25 2026 📍 Groningen, Netherlands

Oxford SPIMAX, Astrophysics department

Seminar 📅 Dec 4 2024 📍 Oxford, UK

Multi-messenger Gravitational Lensing, Royal Society

Invited 📅 Mar 12 2024 📍 Manchester, UK

National Institute for Radio Astronomy (NCRA)

Colloquium 📅 Nov 3 2023 📍 Pune, India

Tata Institute of Fundamental Research (TIFR)

Colloquium 📅 Nov 1 2023 📍 Mumbai, India

SPINS-UK

Seminar 📅 May 5 2023 📍 SPINS-UK seminar (online)

Magnetar telecon (NYU)

Seminar 📅 March 1 2022 📍 online

Cahill Center for Astronomy and Astrophysics (Caltech)

Seminar 📅 Jan 31 2022 📍 online

European Astronomical Society Meeting (EAS 2021)

Invited 📅 July 2 2021 📍 online

CIERA, Northwestern University

Seminar 📅 April 19 2021 📍 online

ASTRON Astrolunch

Seminar 📅 March 3 2021 📍 ASTRON (online)